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Towards Education 4.0: The role of Large Language Models as virtual tutors in chemical engineering

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ABSTRACT

Recent years have seen the rise of Artificial Intelligence (AI) powered generative chatbots, such as OpenAI's ChatGPT or Microsoft's Copilot. These tools have simultaneously positively surprised and taken aback people worldwide, raising the question of whether they can or should be used in education, as well as how to properly guide students and teachers on using them safely and ethically. To this end, this work provides (i) a brief overview of the current applications of AI in Higher Education (HE), (ii) a discussion of the ethical and societal concerns associated with the usage of AI models, (iii) the initial steps of the implementation of a chatbot used at the Technical University of Denmark (DTU) able to perform audits for Good Manufacturing Practice (GMP), and (iv) an investigation of the need and opportunities of AI in chemical engineering education. The latter is achieved through quantitative and qualitative analyses of the responses given by both Master's students and academia/industry practitioners on the introduction and use of AI in education. This paves the way for discussing current perceptions, expectations and concerns of AI models, as well as their limitations and the opportunities they could provide.

1. Introduction

The appeal of using Artificial Intelligence (AI) in university teaching is its potential to meet the needs of both educators and learners. AI in Education (AIED) technologies could provide assistance in a wide range of formal educational contexts, such as through personalized guidance, support and feedback to students, and assist teachers in decision-making (Hwang et al., 2020). According to Ayeni et al. (2024), integrating personalized learning with AI has demonstrated a transformative impact on student performance and academic achievement. Research indicates that it leads to increased student engagement, motivation, and a deeper understanding of the material. AI can in fact contribute to improved retention and comprehension, ultimately translating into enhanced academic outcomes, since students are empowered to take ownership of their learning journey, fostering a sense of self-efficacy and autonomy. In Jian (2023), students subjected to AI-driven personalized learning showed a statistically significant improvement in their grades compared to those in the traditional learning group. These students were more consistent in their performance, suggesting that AI tools provided them a more steady learning curve.

Among the benefits of AI for people, according to the European Parliament, is the fact that it can facilitate access to information, education, and training.[1] Emerging research in the field suggests that the use of AIED has the potential to support both teaching and learning, and improve student performance; however, its misuse, which may be the result of algorithmic bias and the lack of clear regulations, could inhibit human rights and result in the reinforcement of existing inequalities (Prinsloo, 2020; Yang et al., 2021). Despite the expectations and promise of substantial contribution to advancing education, it must be remembered that AI is not infallible, and attention should be paid into its implementation to avoid unintended consequences. Therefore, developing these solutions requires a thorough understanding of the possible issues and limitations associated with AIED. One of the main challenges regarding the adoption of AI is the lack of transparency and accountability and to determine *"who is responsible for the decisions made by the algorithms"*. Responsible AI is a field of research that investigates human

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[1] European Parliament, Artificial intelligence: threats and opportunities, available at: https://www.europarl.europa.eu/news/en/headlines/society/20200918STO87404/artificial-intelligence-threats-and-opportunities.

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responsibility for the development and deployment of AI systems that align with fundamental human values and principles to safeguard human well-being (Dignum, 2019). This responsibility includes various aspects of the AI lifecycle, including development, data acquisition, model implementation and deployment of systems, as well as how end-users understand and adopt the system. Therefore, before designing, implementing and deploying AIED systems, researchers and educators must consider various aspects of equity, responsibility, fairness, autonomy, accountability, and transparency, as well as possible concerns that the system might incite (Prinsloo, 2020; Slade and Prinsloo, 2013; Cerratto Pargman and McGrath, 2021; Velander et al., 2021). Progression in digitalization often stems from a need of educators, for example to save time or automate repetitive processes, rather than from a thorough and unbiased evaluation of the advantages and disadvantages, as well as a deep discussion of how it could impact the end-user. When working in the education field, an assessment of the impact that AI could potentially have on students should be performed. This implies that the learning objectives should be adjusted accordingly and a rigorous evaluation of whether the change to the curriculum and the means used in it is beneficial to the overall learning of the students needs to be in place.

This work seeks to narrow the gap between AIED potential and concerns by providing a student-centric investigation of AI in education through quantitative and qualitative analyses of survey data. One of the objectives is to answer the question *"Are we ready for Large Language Models (LLMs) in education?"* both based on the survey responses, but also by painting a more comprehensive picture of the current challenges and opportunities of integrating AI models into education. This question might seem trivial and futile, since it could be argued that the "AI Revolution" has already started and cannot be stopped. However, it is important to reflect on whether we have taken all the necessary precautions (e.g., through a detailed ethical assessment of what kind of data is stored, how it is used and how could that impact students) and whether we have the needed knowledge for AI, specifically LLMs, to be integrated in the curriculum without hindering the students and their learning. The manuscript is divided into three parts:

- Part 1 focuses on the current application of AIED in Higher Education (HE), both overall and specifically in chemical engineering.
- Part 2 discusses quantitative and qualitative results obtained from two surveys administered to both students and academia/industry practitioners to better understand their perceptions, expectations, concerns, and ultimately if chemical engineers are open and ready for LLM-based virtual tutors (VTs) in education.
- Part 3 draws some conclusions from the gathered data regarding the readiness of the field for the "AI Revolution". It also provides reflections as well as an empirical example of how to safely and ethically develop and deploy AIED systems in the curriculum.

2. Background

2.1. Education 4.0

Education 4.0 represents a significant shift in the education sector due to the rise of Industry 4.0 and digitalization. However, this further brings about the shortage of qualified workforce in particular with regards to successfully driving this initiative forward (Chakraborty et al., 2023). Although there is still no consensus on the definition of Education 4.0 and no standard pedagogical protocol (Chakraborty et al., 2023; Das et al., 2021), it is fair to state that under this educational paradigm, the student is the active agent of the learning process and the teacher mostly provides guidance (Moraes et al., 2023). Education 4.0 targets interactive and personalized learning experiences that promote digital literacy, critical thinking, problem-solving, and creativity, all of which are essential for success in today's workforce. It leverages and prepares the student for the successful and practical use of cutting-edge technologies, such as virtual reality, artificial intelligence and machine learning, data analytics, additive manufacturing, and online platforms, among others. In addition, it aims to produce tech-savvy professionals who can adapt to rapidly changing landscapes and drive progress in this new industrial reality. Hence, its primary objective is to equip students and practitioners with the skills necessary to thrive in a digital and automated world. In this way, Education 4.0 is nothing more than the pathway to acquiring Industry's 4.0 knowledge and the key skill-set (Hariharasudan and Kot, 2018; Moraes et al., 2023).

2.2. AI in education

Pedagogical theories suggest that differentiated and individualized learning is very effective, and therefore educators have been prioritizing this approach across domains (Landrum and McDuffie, 2010; Hwang, 2014). Immediate and continuous feedback has also been shown to be one of the most effective learning methods (Laurillard, 2009; Hattie and Timperley, 2007) as it can prevent small errors and misunderstandings from propagating. AI-driven educational tools are attempting to replicate the same process. This can be done, for example, by using algorithms to learn the preparation level of the students and adapt the learning and exercises accordingly (Lu et al., 2021). It follows that AI is becoming an increasingly important part of new educational platforms in an effort to successfully deploy Education 4.0.

When discussing the introduction of AI in research and teaching, many concerns may arise, such as whether it will replace teachers and scientists or if all aspects of education will be automated, from lectures to exam grading. Hocky and White (2022) conclude that these models have not reduced the need for scientists over time but rather expanded the complexity of the problems that can be tackled. Additionally, the study foresees that these models will increase the accessibility to software tools and greatly increase the scope of what a research group can accomplish. Following these trends, as well as the incessant developments on new and more powerful AI models, research on AIED is also ever-expanding and diverse. Hwang et al. (2020) identified four roles to categorize AIED research: (i) intelligent tutor, (ii) intelligent tutee, (iii) intelligent learning tool or partner, and (iv) policy-making advisors. Prior research has tried to systematically review the use of AI in education, such as in Chen et al. (2020). During the last years, Natural Language Processing (NLP) systems have gained considerable interest in educational research and practice due to their ability to learn and produce natural language. Among others, these models have been used in applications for essay scoring, discourse analysis (e.g., between students), intelligent tutoring systems, and tools that support collaborative learning activities (Ferreira-Mello et al., 2019). NLP models can also be trained to evaluate the coherence of student writing in terms of composition and grammatical correctness (Ramesh and Sanampudi, 2022). An example of NLP-based AIED is provided by RadarMath (Lu et al., 2021), an intelligent tutoring system to support personalized learning for math education. Hwang and Chang (2023), Okonkwo and Ade-Ibijola (2021) and Wollny et al. (2021) delve into the application of Chatbots in education before the rise of LLMs. Another example on an NLP based system is provided by de Las Heras et al. (2020), where a chatbot acts as a peer inside a collaborative e-learning framework.

2.2.1. Large Language Models

Recent years have witnessed the rise of LLMs, which are NLP models trained on massive amounts of text, mostly available online (Brown et al., 2020). This large scale training results in models capable of identifying statistical patterns in massive datasets of content that was originally generated by humans. These models obtain knowledge about a wide range of topics and, hence, are suitable for being applied in many different contexts and fields. A good overview of the main LLMs used is provided by Zhao et al. (2023). This ability to extract content and key information from text offers a powerful tool to enhance the learning experience. According to Huang et al. (2022), these AI tools herald a new era in education, offering unparalleled potential to revolutionize learning experiences and outcomes. They could be used to identify knowledge gaps, assist formative assessment, and provide personalized feedback (Hopfenbeck, 2023; Dai et al., 2023). For example, Wang et al. (2023) presented a large annotated dataset on student insights gathered from available YouTube transcripts, a framework for categorizing feedback types using qualitative analysis, and a set of best practices for using LLMs to cheaply classify student feedback from comments at scale. According to Kasneci et al. (2023), LLMs can also facilitate the evaluation process, resulting in a reduction in teacher workload and providing more accurate and consistent evaluations. Labadze et al. (2023) offers a more recent review of chatbots used in education, including examples of LLMs. Abedi et al. (2023) reviews the use of LLMs in graduate engineering education, discussing the ability to effectively answer complex questions and the advantages of LLMs in the classroom, such as the promotion of self-paced learning, the provision of instantaneous feedback and the reduction of instructors' workload.

Researchers have also focused on evaluating how LLMs perform in a real educational context. Hicke et al. (2023) assess the efficacy of LLMs in generating accurate teacher responses. Xiao et al. (2023) applied LLMs to real-world classroom settings, implementing a reading comprehension exercise generation system that provides high-quality and personalized reading materials to students. They propose extensive evaluations and demonstrate that the system-generated materials are suitable for students and even surpass the quality of existing human-written ones. These strategies could be applied to education across many domains, including chemical engineering.

2.2.2. LLMs in chemical engineering education

According to White (2023), LLMs have been approaching human-level ability across many expert domains. Moreover, they have demonstrated the ability to perform complex tasks in chemistry purely from English instructions, which, they suggest, may transform the future of chemistry and chemical engineering. The potential benefits of integrating LLMs into chemical engineering education are highlighted in Tsai et al. (2023), where ChatGPT (based on the GPT-3 (Brown et al., 2020) and later GPT-4 series of models) is used as a problem-solving tool. In this study, the students had to explore the possible use of LLMs in solving various aspects of chemical engineering problems, specifically applied to the efficiency of the steam turbine cycle. Kong et al. (2023) investigate how ChatGPT can be utilized as an auxiliary tool to create interactive learning environments and simulate real-world engineering thinking processes applied to distillation column design in undergraduate mass-transfer courses. Both studies emphasize the need for students to develop critical thinking skills and a thorough understanding of LLM principles. They claim that this can help improve problem solving abilities and facilitate a deeper understanding of core subjects (Tsai et al., 2023; Kong et al., 2023). Caccavale et al. (2024) presents an initial investigation on how LLMs can be used to automate educational processes through the introduction of FermentAI, a model trained to answer questions about fermentation. However, in spite of these efforts, to the best of our knowledge, LLMs in chemical engineering education are not (yet) vastly widespread. This implies that there is still much research that can be performed in order to establish and disseminate ethical guidelines on how to properly use LLMs in chemical engineering education.

3. Virtual tutors towards Education 4.0

This section presents an empirical study on AI models in education. It aims to provide a student-centric perspective through quantitative and qualitative analyses. Here, we refer to an LLM in the form of a chatbot as "*Virtual Tutor*" (VT). The first part introduces a real use case where ethically sensible AI models could be applied. The purpose of this initiative is two-fold: (i) to improve the learning experience of chemical engineering students and increase their engagement by providing stimulating material and gamification, and (ii) to allow for substantial time savings for the teachers. The second section summarizes opinions and feedback on the application of AIED gathered from two target groups, aiming to compare students' perspectives to industry and academic practitioners.

3.1. From physical to virtual audit

The Department of Chemical and Biochemical Engineering at the Technical University of Denmark (DTU) offers a course on Good Manufacturing Practice (GMP).[2] According to the World Health Organization, GMP prevents errors that cannot be eliminated by quality control of the finished product, and without it, it would be impossible to be sure that all units of a medicine are of the same quality as the units of medicine tested in the laboratory.[3] As part of the course, students are required to participate in an audit exercise, where they assume the role of an auditor investigating a company, represented by teachers, about their good manufacturing practices. During the audit, students can ask questions about the Quality Management System, the available documents and practices adopted by the company, which is necessary information to evaluate whether the company in question could be a viable business partner or if detected non-conformities are too severe. The role of teachers is to provide accurate answers in case the documents are available, or vague responses, leaving students to reflect on the company's behavior. After performing the audit exercise, a report of the findings must be handed in. Notably, this report is not the final exam that students are graded on, but an activity that needs to be of sufficient quality. The report is then evaluated in order to assess the groups' learnings from the audit exercise and their position towards the company's good manufacturing practices.

After a few years of participating in the audit exercise, teachers have agreed to investigate the replacement of physical teachers with an AI-powered VT. The course is generally quite discursive, and the regulations and documentation are straightforward to discuss, so it poses a suitable use case for the application of LLMs. The hypothesis behind introducing AI to the course is that students will be more engaged by the gamified nature of the VT and will not be afraid or reluctant to ask for further questions. This work presents the early stages of this experiment, with the objective of gathering students' and other chemical engineers' perspectives on the development of a digital audit tool to represent the fictional company and, therefore, replace the teachers in repetitive tasks.

To be noted is that this initiative does not aim to move towards a teacher-less course, but rather to help teachers focus on the lectures and material preparation instead of exercises that are very time-consuming and ultimately quite repetitive. In fact, each year the 3 teachers responsible for the course have to allot the audits of 24 groups of 5–6 students. This imposes a physical limitation on the annual course uptake, where approximately 25 % of enrollments are rejected. Moreover, an argument could be made that the enthusiasm of teachers diminishes throughout the audits. Therefore, the objective of this work is to automate a repetitive task, and this specific course provides a perfect candidate for this experiment, given that the questions asked by the groups of students always cover the same topics and are generally quite similar (both syntactically and semantically).

[2] 28855 Good Manufacturing Practice (GMP) and quality in pharmaceutical, biotech and food industry - Theoretical version, available at: https://kurser.dtu.dk/course/2023–2024/28855.

[3] Medicines: Good manufacturing practices, available at: https://www.who.int/news-room/questions-and-answers/item/medicines-good-manufacturing-processes.

3.2. Students and practitioners' perspectives

3.2.1. Selected targets

To evaluate the introduction of virtual tutors in education, we conducted two different surveys: one aimed at students enrolled in the GMP course and one for participants in the *European Network for Business and Industrial Statistics (ENBIS) Spring Meeting on Digital Twins* [4].

The first survey was completed by 22 students who were enrolled in the GMP course in the spring of 2023 and the second by 12 participants attending the ENBIS conference in May 2023. The choice of the first group is motivated by the fact that students attending the GMP course will be the target of the experiment. Therefore, we deemed it necessary to assess their openness and readiness when it relates to a change that will affect them directly. The second group was selected due to its variety, representing well the diversity of occupations within chemical engineering, from academia (PhD students, postdocs, professors) to industry practitioners (graduates, seniors, managers). Further details on the occupation and rank of these respondents were not collected; however, they may include responses from researchers actively working on integrating AI into education that could be regarded as expert opinions.

Furthermore, the aim of gathering responses from different target audiences was to have a broader view on how AI is perceived in education, both from students, who will be directly affected by its introduction, and from academia/industry practitioners, who will only indirectly witness the effects of using AI in the educational system.

Both surveys contain purposely thought-provoking questions, such as *"Do you think AI tutors should replace physical teachers?"* to evaluate the extent to which respondents agree with AI to completely replace teachers. Here, we were mostly interested in their arguments supporting the given answers, since both surveys contained many open questions where the participants were asked to justify their opinions. This led to interesting takes on AI in education, including fears, hopes, and doubts of people not strictly working with it, but that will likely be subjected to its possible benefits and repercussions. All participants in the surveys were informed that the responses would be used in research and published upon anonymization. They were informed that no emails or other personal information would be stored or used at any time.

3.2.2. Survey results

One of the surveys was administered to students attending the GMP course during the spring semester of 2023. These students participated in the physical audit and generally had a very positive experience with it. In fact, in the given survey, to the question "On a scale from 1 to 5, how helpful do you think the audit exercise was for you to learn GMP concepts?", 36.4 % responded with a 4, and 63.6 % with a 5. Additionally, Fig. 1 shows students' perceptions regarding the responses provided by the teachers during the in-person audit exercise. The objective of the investigation performed was specifically to gather students' opinions on the possibility of AI agents replacing teachers in the audit exercise. From the survey, we observe that, in general, most students are unsure about replacing teachers with VTs. Fig. 2 (a) shows that most students (68.2 %) answered "Maybe" to whether they think that a VT should replace teachers, specifically in the GMP course. Among the rest, 22.7 % is against this, and 9.1 % favors a VT to perform the audit. In particular, when asked about the general domain (that is, if VTs should replace teachers not just in this course but in general in the educational system), a much higher percentage of students are against it (36. 4 %), only 4. 5 % is strictly in favor, and the rest believe that VT should support teachers.

Moreover, both groups were asked to elaborate on the answer. Their answers, shown in Table 1, convey that they doubt the ability of the LLMs to understand *"connotations of words"* and *"complex formulations"*, and they highlight the perceived limitations of current models (i.e., *"I think at the moment they support the teacher as they are quite unreliable"*). Specifically for the GMP course, some students think that the interaction with the teachers is enriching and instructive and, therefore, are hesitant towards a VT (e.g., a chatbot) performing the same task. In general, some responses emphasize the idea that *"there is a lot of value in human-to-human tutoring and learning, and therefore [replacing teachers with AI tutors] would be a definite loss of quality in teaching"*. Moreover, some students are afraid of the possibility that teachers will lose their jobs and that AI systems lack pedagogical competencies. However, both students and ENBIS participants suggest valuable applications where VTs could be employed, for example, for *"reducing their [the teachers] workload, being more available to the students, and also potentially improving the overall quality of the information given to the student"*.

Some of the opinions are very favorable regarding AI models being used in education. Among the students, some are curious and optimistic about these models being employed in the course provided the right setup (e.g., *"If the chatbot will have all the necessary documents then why not"*), and highlight the repetitive nature of the task and its suitability for AI applications, e.g., *"Usually the same questions are asked by the students, which is ideal for a chatbot"*. Even applied to the general domain, some students have positive opinions of using VTs in education and think that having VTs would be good *"if it can relieve some of the work from the teachers, so it will increase the quality of the teaching [...]."*. Finally, among conference attendees, someone suggested that AI tutors *"are inevitably going to be part of the future, so it is important to implement it"*. After gathering general (positive and negative) opinions regarding AI in education, we specifically focus on addressing the responders' concerns. Fig. 2 (b) investigates whether the students had any concerns about VTs replacing teachers in the GMP course. The results show that most of the students (16/22) have concerns about the quality of the responses that a VT could generate. Noteworthy is that students are not strictly against VTs (i.e., only 2/22 students don't think that a VT could perform the audit) but rather focus on the overall outcome quality. The students then elaborated on their opinions in a follow-up question. For further details, their responses are summarized in Table 3, which can be found in Appendix A.

Students attending the GMP course were also asked to answer when they considered it better to have a physical teacher and when they thought a virtual tutor would be more beneficial. Selected responses are presented in Table 2, while the full table can be found in Appendix A. The responses show that some students think having a physical teacher is always better, while others see the potential benefits of having virtual tutors in certain cases, such as for *"questions to the lecture, which [could be] easily answered"*, or *"when the student needs fast answers"*. They also highlight that there should be *"a lengthy trial period as all this AI/chatbot based learning is quite a new concept"*, which is an important reflection for educators to take into account. Most students agree that physical teachers are preferable to virtual tutors for lectures to *"add personal experiences as a part of the related subject"* and *"for courses that require more experimental activities"*.

In addition, both groups were asked their opinions about the medium they prefer for the virtual tutor. Students attending the course seem to favor an agent (human-like tutor), followed by a chatbot (text-based), while conference participants favor a chatbot, followed by an avatar (more cartooned version). Among the reasons behind the choice of an agent, responders enumerate the interactive nature, the fact that the communication would feel more real and quick (no need to type the questions), and to simulate a situation they are familiar with. Chatbots seemed to be preferable since they could provide simple and quick clarifications, they could be saved for later consultation, and it could

[4] The ENBIS conference was hosted in Copenhagen on May 25–26, further details are available at the following link: https://conferences.enbis.org/event/35/.

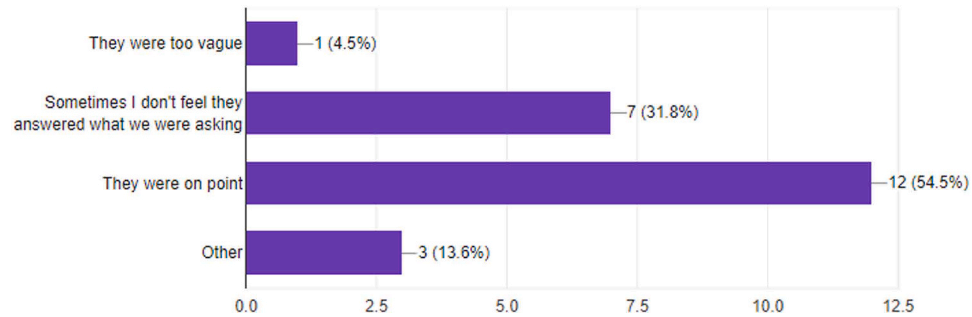

Fig. 1. Responses of the students to the question: *"What do you think about the teachers' responses?"*. This question refers to the experience of the students during the physical audit exercise performed with the teachers.

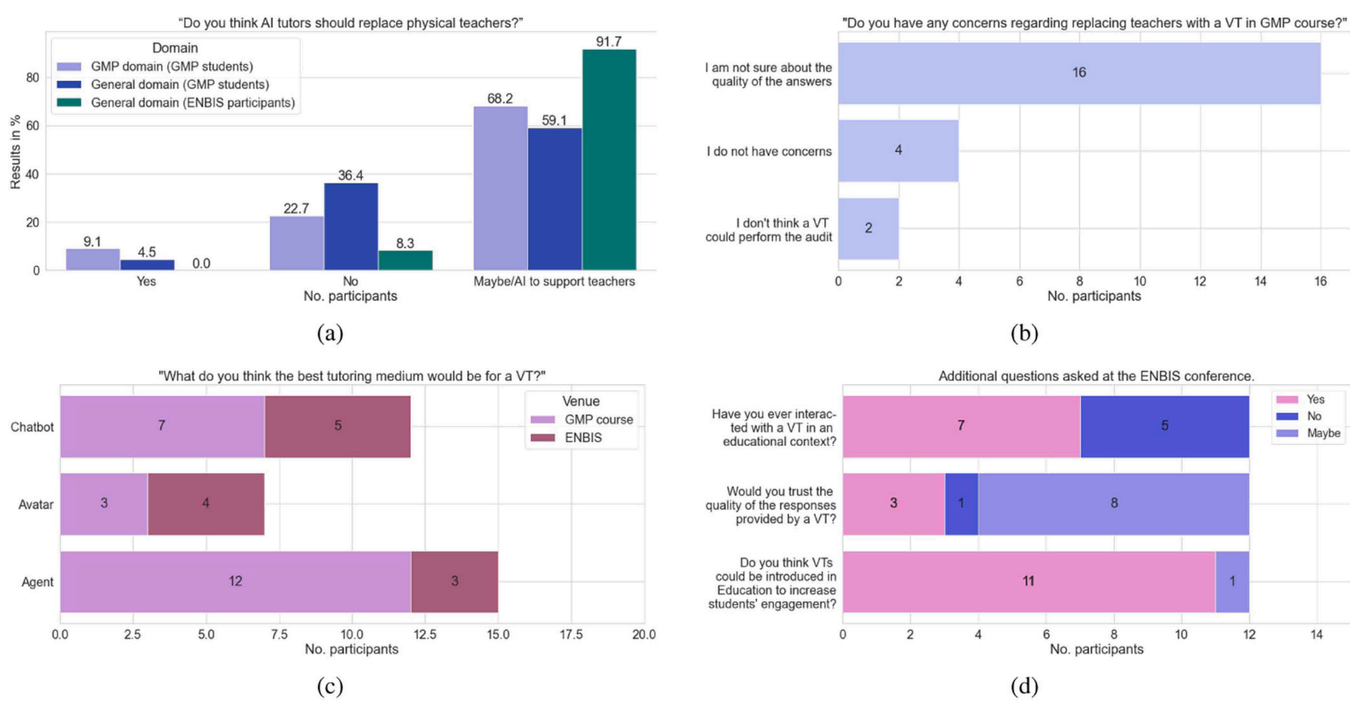

Fig. 2. (a) Response to the question *"Do you think AI tutors should replace physical teachers?"* both in the GMP course and in the general domain. The first two bars are responses from students enrolled in the course (22 students), and the third bar shows data collected during the ENBIS conference (12 participants). (b) the response of the GMP students to the question *"Do you have any concerns about replacing teachers with a virtual tutor in the GMP course?"*. (c) Response to the question *"What do you think the best tutoring medium would be?"* of GMP students (left) and ENBIS participants (right). (d) Multiple questions were asked at the ENBIS conference.

prevent the uncanny valley effect[5] of having a VT that seemed "too real". Finally, for avatars, some responders mention that *"human-like can be scary and a chatbot is a bit too boring"*. For further details, the participants' responses regarding the chosen medium are summarized in Appendix A in Table 5.

ENBIS participants were asked to answer some additional questions, shown in Fig. 2 (d). Their responses to the question *"Would you trust the quality of the responses provided by a VT?"* are consistent with the students' perceptions shown in Fig. 2 (b). The majority thinks that VTs could be introduced in education to increase students' engagement.

4. Discussion

The work presented in this manuscript aims to help researchers and educators assess and reflect on the possible impact of AIED systems and whether we are ready to embrace the shift. In this section, we discuss possible challenges and opportunities of deploying AI models in educational settings. We especially focus on the case of chatbots, introducing an initial investigation conducted at DTU. We also argue for a more collaborative and user-centric perspective when considering if and how to use these models.

4.1. Current guidelines on AI systems in EU

The focus on ethics and fairness in AI has gained traction in recent years, both by researchers (e.g., Holmes et al., 2021; Slade and Prinsloo, 2013; Cerratto Pargman and McGrath, 2021; Yang et al., 2021) and governmental institutions such as UNESCO and the European Parliament.

In recent years, extensive research has been conducted on the opportunities and challenges imposed by the use of AI in education. In fact, although AI has proven to be beneficial in many aspects, such as

[5] Definition of Uncanny Valley by Britannica, available at the following link: https://www.britannica.com/topic/uncanny-valley.

Table 1
Selection of the top 10 students' opinions regarding virtual tutors, both in the course they are enrolled in GMP and in the general domain. The order of the responses is randomized (not the same student's answers in the same row).

Q: Do you think the teachers' role could be replaced by a VT in the course?	Q: In general, do you think AI tutors should replace physical teachers? (GMP course)	Q: In general, do you think AI tutors should replace physical teachers? (ENBIS)
"It would save time and we can obtain more accurate answers. Although, an experienced teacher can also fulfill his role and allow the students to learn from the activity"	"Although the material that teachers present and teach can be quite similar from semester to semester and year to year, the experience of teaching and being taught varies depending on the audience. And that is valuable."	"AI can enhance bit of course is inherently flawed due to the natural constraints based off the algorithm and training data. It cannot as of yet, replace human teachers but can enhance productivity."
"Depends how good it is"	"In a far future, it is a very high possibility"	"We don't live in a metaverse. "
"Some questions could be answered by a bot (fx: the bot could provide SOPs when asked), but I doubt they can provide correct answers if the questions are formulated a bit complex."	"I think there is a lot of value in human to human tutoring and learning, and therefore I think it would be a definite loss of quality in the teaching. [...] They could support the teacher, as some students would likely have no problem navigating [...]. But it might be a slippery slope as they are likely cheaper, and other students would likely fail to capture the strengths."	"I think there is still value in the human interaction, and the AI may not be able to capture all the nuances of a subject. Therefore I believe they should instead support the teacher. Reducing their workload, being more available to the students and also potentially improving the overall quality of the information given to the student."
"21st century, the future is now"	"The experience of the teachers play a big role, so I think currently AI should only work as a supporting role, not completely replacing the teachers"	"Like chatGPT there has to be someone that controls the answers given."
"The lack of humanity could represent a problem, and a chatbot will not understand connotations of words"	"I think it's important to have human mentor to feel connected to a subject and more motivated as 'why should I even learn this, if bot can do it all for me?"'	"I think at the moment they support the teacher as they are quite unreliable."
"For me it is important the interaction between teacher-alumni, it cannot be replaced by a virtual tutor	Again, the social aspect is extremely important. A generation of people who only socially interact with a chat bot is doomed"	"They probably can never replace but they more they can support, the more the teachers can use resources on research."
"Finding non-conformities is a lot about digging into certain topics. If the company knows they are in the wrong, why would the tutor admit it?"	"If bots replaced teachers, we might as well not show up for lectures and just study directly form the text books. Bots can support by doing simple-minded tasks."	"Should work as a support to the teacher, the teacher has experience and knowledge which may not be transferred through AI."
"Many of the questions would be easy with a chatbot, however, if a question is asked that the chatbot does not know the answer to, it can not come up with an answer on the spot."	"If it can relieve some of the work from the teachers, so it will increase the quality of the teaching, it would be good."	"I see this as a clever frequently asked questions section on a webpage."
"If the chatbot will have all the necessary documents then why not"	"I think human interaction is what makes lectures special."	"They're inevitably going to be a part of the future so it is important to implement it."
"Usually the same questions are asked by the students, which is ideal for a chatbot"	"I don't want the teachers to lose their job, plus AI tutors might be factual, but they lack pedagogical skills"	"Asking questions becomes much easier and cuts waiting time which is big problem for engagement, but human interaction should not be removed."

Table 2
Selection of the top 10 students' opinions regarding preference between physical and virtual teachers. The order of the responses reflects the opinions of each student (each row corresponds to a unique student).

Q: When do you think it would be better to have a physical teacher?	Q: When do you think it would be better to have a virtual tutor?
"Every time."	"Never really."
"During the lectures and exercises (especially during the beginning of the course)."	"During study for the exam, when the student needs fast answers."
"At the lectures it is better to have physical teachers to present."	"When you need help with an assignment or have any questions."
"I would say lecture should be with physical teachers, where they can add personal experiences as a part of related subject."	"Virtual tutor will be better in like strictly theoretical parts, relating on laws, rules."
"For courses that require more experimental activities."	"For some theoretical courses."
"Helping with tasks and assignments - and teaching."	"Questions to the lecture, which is easily answered."
"At most times."	"I think physical help should always be provided, but a virtual tutor could play a role in supporting. Under all circumstances should there be a lengthy trial period as all this AI/chatbot based learning is quite a new concept."
"For everything."	"For non accessible hours for the teachers."
"Normal lectures, exercises, exams,... always; never replace a physical teacher."	"Never; maybe just in exercises helping depending on their methodology."
"Mostly all the time."	"They could help to explain, for example, the correct solutions for the practice problems and the reasoning behind them, or can give helpful tips for the audit preparation."

accuracy of predictions and the ability to solve complex problems, it can pose risks to privacy and lead to discrimination, as highlighted by the EU guidelines on Ethics in Artificial Intelligence document,[6] published in 2019. This document provides a good overview of the possible ethical concerns surrounding AI and the principles that must be followed to ensure its ethical applications. The core principle of the EU guidelines is that the EU must develop a 'human-centric' approach to AI that is respectful of European values and principles. This approach should strive to ensure that human values are central to the way AI systems are developed, deployed, used and monitored, by ensuring respect for fundamental rights, including those established in the Treaties of the EU and the Charter of Fundamental Rights of the EU. All of these principles

[6] EU guidelines on ethics in AI: Context and implementation (2019), available at: https://www.europarl.europa.eu/RegData/etudes/BRIE/2019/640163/EPRS_BRI(2019)640163_EN.pdf.

are united by reference to a common foundation rooted in respect for human dignity, in which the human being enjoys a unique and inalienable moral status (Caccavale et al., 2022). On 24 November 2021, UNESCO's General Conference adopted the Recommendation on the Ethics of Artificial Intelligence.[7] This work builds on the preliminary study on the Ethics of AI[8] of UNESCO's World Commission on the Ethics of Scientific Knowledge and Technology (COMEST).[9] The core of this study is that no global instrument covers all the fields that guide the development and application of AI in a human-centered approach. In 2022, two documents were published aimed at addressing AI in education: the EU Commission published guidelines to help teachers address misconceptions about AI and promote its ethical use,[10] and The European Council published the "Artificial Intelligence and Education: A critical view through the lens of human rights, democracy and the rule of law".[11]

Although these efforts are laudable, the current guidelines are nonbinding. However, concerns have been raised about the lack of regulatory oversight to support their implementation. In April 2021, the EU Commission put forward its proposal for the AI ACT, the world's first comprehensive AI law, to ensure better conditions for the development and use of AI. The EU Parliament wants all AI systems used in the EU to be *"safe, transparent, traceable, nondiscriminatory, and environmentally friendly (as also tackled in* (Bender et al., 2021). *AI systems should be overseen by people, rather than by automation, to prevent harmful outcomes"* [12]. On 14 June 2023, the Members of the European Parliament adopted the Parliament's negotiating position on the AI Act.[13] On 8 December 2023, the Council presidency and the European Parliament negotiators reached a provisional agreement on the proposal on rules regarding AI, the so-called Artificial Intelligence Act. The draft regulation aims to ensure that AI systems placed on the European market and used in the EU are safe and respect fundamental rights and EU values. The core of this initiative is to regulate AI based on its capacity to cause harm to society following a 'risk-based' approach: the higher the risk, the stricter the rules.[14] In this agreement, the use of AI in an educational context is classified as high-risk, and therefore these AI systems will have to be registered in an EU database.[15] After the formulation of the EU AI Act, the agreed text has to be formally adopted by both Parliament and Council to become EU law. It was approved by the Council on 24 May 2024.

4.2. Impact of introducing AI in education

Given the potential concerns and the lack of strict common regulations worldwide, it is very important to reflect on *"how to ethically and fairly employ AI models in education"* and do this ethically and systematically. These systems, if used incorrectly or deployed too hurriedly to accommodate the fast-growing demand to digitalize education and stay on par with times and technologies, could cause irreparable damage to students, such as violating their privacy or providing erroneous information that could be then propagated throughout their studies. Considering how data is acquired, stored, and used is of great importance. In fact, data is considered a valuable resource, since it could provide extensive knowledge about personal habits, preferences, and skills. Thus, it could be used for economic gains outside of its primary scope (e.g., for targeted advertisements) if not regulated. We believe that researchers working on the design and implementation of AI models should be familiar with ethical principles, and ethics should be taught in technical degrees. However, since evaluating possible ethical implications would likely require the cooperation of stakeholders of varying levels of expertise, regulations must be established and enforced at a higher level (Raji et al., 2021). The ethics crisis does not only affect the people directly working on AI models and, therefore, should not be solved only by technical people, but as a collaborative effort among different disciplines. Echoing Raji et al. (2021), we argue that solutions and guidelines to address the ethics crisis cannot be siloed, that is, each discipline working on a solution without considering other points of view. The best way to address this issue, which should be perceived as a transversal problem that involves theories and methods in several disciplines, is through a collaborative effort between disciplines. For further arguments on the matter, see Caccavale et al. (2022). Finally, the current guidelines and regulations are very region-specific, while we believe that global laws on the use of AI in various fields, including education, should be stipulated.

Another possible concern that researchers must account for before using AI models in education is robustness. As also discussed by Huang et al. (2022), these models still present some drawbacks, such as technological limitation, novelty effects, and cognitive load during the learning process of the students. Teachers must thoroughly investigate these aspects before deploying these models in the educational system. One of the main concerns highlighted in the survey was the quality of the generation produced by a VT. Despite the initial skepticism provoked by the responses given by generative chatbots, also highlighted in some of the survey data presented, the latest improvements in these technologies are proving that LLMs are quickly becoming better and better. However, before introducing these models in educational systems, extensive testing and validation (of the model, interface, novelty introduced and cognitive load) must be performed.

Although an AIED could be used fairly and ethically, we should also reflect on whether these systems could benefit students and improve their education on a case-by-case basis and whether they are not just implemented as an expedient to save time and money. LLMs and other AI models are powerful and can handle a wide variety of tasks, but it is important to remember that humans are still in charge of making the decision of when to use these models and who benefits from them. In Section 3.2, we present the perspectives of two different target groups on the use of AI models in education. It is also important to take these opinions into account when considering the use of AI in an educational context and, if possible, ask students what their thoughts are on the matter, since they will be directly affected by it. Among some of these opinions, students expressed the thought that using AI in education could be beneficial because it would lead to reducing teachers workload and them being more available to the students, which could also potentially lead to an improved overall quality of the information given. Moreover, *"asking questions becomes much easier and cuts waiting time which is big problem for engagement, but human interaction should not be removed"*. The interaction between humans and AI is also a crucial

[7] Recommendation on the Ethics of AI, available at: https://en.unesco.org/artificial-intelligence/ethics#recommendation.

[8] Preliminary study on the Ethics of AI (2019), available at: https://unesdoc.unesco.org/ark:/48223/pf0000367823.

[9] COMEST, available at: https://en.unesco.org/themes/ethics-science-and-technology/comest.

[10] EU Commission: Guidelines for teachers, available at: https://ec.europa.eu/commission/presscorner/detail/en/ip_22_6338.

[11] AI and Education, available at: https://rm.coe.int/prems-092922-gbr-2517-ai-and-education-txt-16x24-web/1680a956e3.

[12] EU AI Act: first regulation on AI, available at: https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence.

[13] Parliament positions on AI, available at: https://www.europarl.europa.eu/news/en/press-room/20230609IPR96212/meps-ready-to-negotiate-first-ever-rules-for-safe-and-transparent-ai.

[14] Artificial intelligence act: Council and Parliament strike a deal on the first rules for AI in the world, available at: https://www.consilium.europa.eu/en/press/press-releases/2023/12/09/artificial-intelligence-act-council-and-parliament-strike-a-deal-on-the-first-worldwide-rules-for-ai/.

[15] EU AI Act: first regulation on artificial intelligence, available at: https://www.europarl.europa.eu/news/en/headlines/society/20230601STO93804/eu-ai-act-first-regulation-on-artificial-intelligence.

aspect to consider when developing AI solutions in any field, including education. In fact, if a tool is provided through a sub-optimal interface, it could lead to frustration and ultimately students not being willing to use it, and therefore not fulfilling the learning objectives.

Finally, as with every major transformation, new skills will be required in the future, and educators should consider *what the students should learn in this early age of AI*.

4.3. Are we ready for LLMs in education?

New and state-of-the-art generative AI models are constantly being released at an impressively fast pace. This implies that technology is certainly available for these models to be applied in almost any field. However, the regulations and societal sensibility around the matter might in many cases be the limiting factor. These models provide undeniable opportunities, such as improving the quality of teaching and learning, automating repetitive tasks, and helping teachers use their time more efficiently. The survey responses highlighted the openness of the participants toward the use of these systems in an educational context. Remarkably, when asked about a context they are familiar with, students are twice as likely to state that AI tutors should replace teachers compared to when asked about the general domain. When asked the same question, the vast majority of conference participants responded that AI should be more about supporting than replacing teachers, while 8.3 % is against it, and none of the respondents is in favor of completely replacing teachers. Interestingly, it seems that more students than graduates are against the use of AIED systems, perhaps because it will directly impact them and their learning path. On the other hand, people attending the conference seem to focus more on the possible benefits of AI supporting the teaching activities rather than the possible issues and downsides. To be noted is that no one strictly favors AI just replacing teachers. The majority of the respondents believes that AI should support and not replace teachers, as well as the fact that it could be introduced to increase engagement.

The question asked has, however, a faceting nature, and therefore multiple points of view are needed in order to tackle it and answer in a more exhaustive manner. One might argue that initial resistance and skepticism is often present when a new tool or technology is introduced. An example could be provided by the introduction of Python in chemical engineering, compared to the widely used MATLAB. Although its versatility and other attributes, Python is still not fully embraced by the community (Caccavale et al., 2023). More recently, ChatGPT also raised some skepticism (Watters and Lemanski, 2023). Therefore, the question asked might not be entirely the right one. As a respondent stated, AI tutors *"are inevitably going to be part of the future, so it is important to implement [them]"*. Maybe it is not a mere matter of being ready, because if we do not push the progress we will never be, but more a matter of *being ready to take any measure to ensure an ethical and unbiased used of the technology and being ready to consistently monitor the learning of the students to assure that the new methods don't hinder their educational path overall.*

Given that researchers and educators follow the principles discussed to avoid the aforementioned drawbacks, we believe that AI models could be safely embedded in the educational system.

4.4. Our approach

As an example of how to achieve this, after analyzing the responses of the students attending the GMP course taught at DTU and their openness to test an AI chatbot, we have been taking on the challenge and have started working on a VT as a digital audit tool. This work spans over two years, where the majority of time has been spent on data collection and pre-processing, to ensure safe and high-quality data. Throughout this process, we performed data anonymization and other practices to prevent the storage of personal information. Before being used in the course, the model has been extensively validated, both quantitatively and qualitatively. First, it was tested in a small-scale test and then also with experts in the subject (the teachers). This qualitative assessment was conducted to rigorously evaluate the responses of the models and to assess their applicability to the course.

To address some of the issues highlighted above, we propose a prompt engineering-based approach, in which we limit the generative ability of the LLM through the use of more conservative parameters and instructions to favor more accurate and reliable answers. We believe that researchers and educators must address the trade-off between generative power and reliability when AI models are applied in an educational context to ensure ethical responses. This stems from the fact that students might not be as familiar with the subject being taught and, consequently, they might not have the critical ability to discern wrong from correct information. Our responsibility as educators is to ensure, as much as possible, that the information provided to students from AIED systems is reliable and trustworthy. Although many positive points could be made for using generative models without any 'restriction' (such as the impressive nature of the generated output and their "creativity"), we must also ask ourselves whether the responses are correct. Our approach is to explicitly instruct the model to prevent hallucinations, which we believe to be a valid strategy to obtain more reliable answers.

5. Future directions

Our approach will be described in further details in next publications. The LLM-based VT has been tested in the audit exercise in the 2024 edition of the course at DTU. Groups of students were asked to volunteer for the experiment, to avoid the risk of hindering the experience of some students who might not be comfortable switching to a VT yet. We will also include an in-depth discussion of the interaction between humans and AI, where the experiences of students performing the audit exercise with the VT are compared to the regular audits (performed with the physical teacher).

The next work will also contain teachers' and other experts' opinions, as well as the quantitative results of the audit exercise performed with the chatbot. Therefore, we will compare the "naïve" perspective of students before having tried the VT to their actual experience with it (after being trained on the best practices) and the teachers' reflections on the experiment. The current manuscript aims to provide an initial theoretical framework necessary for the next work to build upon.

Finally, to further evaluate the VT, we plan to benchmark various approaches in the future, such as comparing prompt engineering and instruction fine-tuning, to investigate what is the best strategy to adopt in this use case and whether it can be generalized to different applications. For example, fine-tuning could be beneficial in fields that use language in an idiosyncratic manner, such as in the case study presented.

6. Limitations

This study aims to provide a student-centric perspective, as students are the ones to be directly affected by the changes. The focus is deliberately on their opinions and the experience that they believe they would have with an AI tutor. This is in contrast to sharing the opinions of experts from a pedagogical or theoretical perspective.

In one of the surveys (the one rolled out during the conference) we asked whether respondents have actually interacted with a chatbot in an educational context. 58.3 % of respondents have, while 41.7 % did not at the time. It is noteworthy to mention that the survey was administered in May 2023, so ChatGPT was already available but not as widely accepted as now and the opinion were most likely based on GPT-3.5 (unless someone had access to the paid version).

7. Conclusion

This study presented (i) an investigation of the implications of using

AI in education, (ii) quantitative and qualitative data regarding the use of AIED systems, and (iii) a description of the initial stages of a chatbot used to perform an audit exercise in the GMP Master's level course taught at DTU.

We presented quantitative and qualitative responses from students and industry/academia practitioners regarding the introduction of AI in education. In general, both groups agreed that AI models should not replace teachers. Their main concern was the quality of the output generated. However, both groups agreed that AI should be used to support teachers, which could lead to potential time savings and the opportunity for teachers to focus on teaching quality and other important activities.

We believe that this effort will encourage researchers and practitioners working in this field to contribute with reflections and resources to ensure a more ethical and safe transition into the "AI Revolution". This will thus safeguard learners and guarantee that every aspect of the 'digital contract' is transparent.

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CRediT authorship contribution statement

Fiammetta Caccavale: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Carina L. Gargalo:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation. **Krist V. Gernaey:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition. **Ulrich Krühne:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table 3
Students' opinions on the potential concerns of introducing AI in education.

Q: Do you have any concerns about replacing the teachers with a VT in the GMP course?	Q: In general, do you have any concerns regarding virtual tutors?
"[…] Sometimes we don't formulate our questions in the right way and I don't know if the VT would be able to respond well to that."	"No I don't know I think so."
"I feel like the audit is more a conversation and should give a holistic image of the company which a virtual tutor might not be able to do."	"Yes, I think they will quickly learn to find common non-conformities."
"[…] If questions not in the system are asked it can not answer."	"A bot can not come up with answers and explain something differently, unless it is a very advanced bot."
"Any concern at all."	"They depend on the wifi, the internet connection, you can't have interaction at all…"
"Sometimes the answers we received were vague and after some nagging we got the answers during the audit. Not sure if the chatbot can do the same, since the interaction made the difference - but I am open to changing my mind."	"No"
"Yes, maybe the chatbot will be repeating itself?"	"A bit."
"Maybe it would be too impersonal, maybe the bot might have problems understanding questions (e.g., subtitles on audit video are not correct)."	"May affect the will to learn."
"It is hard to say, if we can get the same in-depth answers as the teacher gave us."	"It might not answer questions that are not in frequently asked questions."
"I would be interesting to see how that would work."	"No."
"For us it was good to learn to interrupt the teacher to keep up with the time plan."	"I am afraid that everyone would end up sitting at home like during corona [Covid-19] lock down."
"I believe it will take a lot of time and trials for the chatbot to be able to replace the teachers effectively. Also a big part of the audit consists of the auditors asking improvised questions on the spot. I'm not sure if the chatbot will be able to cope."	"They should be introduced in teaching but not totally replace tutors."
"It should definitely be better than what ChatGPT is now, if the level would be the same, the quality and performance would be low, in my opinion."	"I do not think we should strive to live at home just because we can. What is next if teachers are replaced? My education and future job is pointless then if AI can just do it."
"See previous answer."	"Firing and replacing physical teachers."
"I am not sure about the quality of the answers a virtual tutor could provide."	"No, its interesting."
"The experience will be further away from a real audit, if you don't even sit and talk to a real person about the challenges of their company."	"Not getting elaboration enough on the questions or not getting the answers to the asked questions."
"I think it could be done, but there would be a great loss of nuances, and the practical experience of conducting the audit would likely go completely to waste as there would be no personal aspect to it."	"LOTS."
"The lack of humanity could represent a problem, and a chatbot will not understand connotations of words."	"Yes."
"I think a virtual tutor could give proper answers and it could be useful, but I don't think it is a good idea to replace human people with robots in that case, since the feeling of accompaniment to the students could not be offered."	"Yes, lots of concerns, as I have expressed in the previous answers."
"They can help each other."	"No."
"I think the current virtual solutions are not on the level yet that they could replace a teacher."	"The information they gave might be false, or their evaluations of our virtual audit might be biased."

Table 4
Students' opinions regarding preference between physical and virtual teachers. The order of the responses reflects each student's opinions (each row corresponds to a unique student).

Q: When do you think it would be better to have a physical teacher?	Q: When do you think it would be better to have a virtual tutor?
"For questions about assignment and report especially"	"Maybe to go over certain theoretical things or as a demonstration on a plate site idk"
"Always always always."	"Maybe for group work exercises or GMP training in general.""
"It is better to have a physical teacher when questions are not perfectly predictable"	"When questions are very predictable (often asked questions) a bot will be as good as a person"
"Every time"	"Never really"
"During the lectures and exercises (especially during the beginning of the course)"	"During study for the exam, when the student needs fast answers"
"Important subjects and when information or a guide will be presented face-to-face."	"Assignment guide, information announce and document supply"
"At the lectures it is better to have physical teachers to present"	"When you need help with an assignment or have any questions"
"Help during group work and assignments, discussions in class, the oral exam"	"In creating specific new problems for students to work with, in situations that do not need discussions such as set-in-stone knowledge, getting some feedback on assignments"
"I would say lecture should be with physical teachers, where they can add personal experiences as a part of related subject"	"Virtual tutor will be better in like strictly theoretical parts, relating on laws, rules"
"For courses that require more experimental activities"	"For some theoretical courses"
"When asking questions"	"In the lectures"
"Lectures"	"Interactive tutoring sessions."
"I think a conversation is nice. I might start with one question and ask further questions. I would hate if I had to talk to my computer or write every single question. Too time consuming"	"Less time consuming for teachers"
"During lectures in case you have questions regarding a topic. And when you need help with a task."	"When the teacher is sick and needs a substitute"
"Both of them its good to help each other."	"Both of them its good to help each other."
"Helping with tasks and assignments - and teaching"	"Questions to the lecture, which is easily answered"
"At most times."	"I think physical help should always be provided, but a virtual tutor could play a role in supporting. Under all circumstances should there be a lengthy trial period as all this AI/chatbot based learning is quite a new concept."
"For everything"	"For non accessible hours for the teachers"
"Normal lectures, exercises, exams,… always; never replace a physical teacher."	"Never; maybe just in exercises helping depending on their methodology."
"Mostly all the time"	"They could help to explain, for example, the correct solutions for the practice problems and the reasoning behind them, or can give helpful tips for the audit preparation"

Table 5
Students' opinions regarding the preferred medium to be used.

Best medium	Q: Could you elaborate on your preference?
Agent (human-like)	It feels really weird to have to use a chatbot because then we might as well just receive a detailed description of the company structure and read our way through it instead of having to ask for everything.
Chatbot (text-based)	If the avatar/agent is supposed to use a read out the answer that a chat bot gives i can not see it making sense, as having the writen word would be easier to use.
Agent (human-like)	Because of the interaction, the similar to the human-like, the better
Chatbot (text-based)	Fast and easy clarifications of potential questions
Avatar	Human-like can be scary and chatbot is a bit too boring.
Chatbot (text-based)	I think through texts we can describe the best what are we searching for
Chatbot (text-based)	A chatbot that answers questions (like chatGPT), but also that provides images, documents, etc
Agent (human-like)	You have much more time to think about what you are asking when you write the answers which would make the audit feel less real
Chatbot (text-based)	A real human. Though the chatbot/avatar/whatever solution would be nice, the social interaction between humans is important. Otherwise I could just be at home talking to my computer. More than I do already. And that does not prepare me for a job in the real world
Agent (human-like)	The more human-like the bot is, the more 'real' the audit feels
Chatbot (text-based)	Unless it was very very good it would feel more fake to employ a more human like medium, and therefore I would likely stay with a chat based dialogue.
Chatbot (text-based)	Can save it for later
Agent (human-like)	The same as in the other questions; I don't think the human approach can be replaced, and having a robot/chatbot would make everything much colder.
Agent (human-like)	Feels more like traditional teaching
Chatbot (text-based)	I would be satisfied with a chatbot but some dyslexic people might be a fan of read out loud answers
Chatbot (text-based)	I dont see the need to humanize AI
Chatbot (text-based)	The ease of use is very important
Agent (human-like)	To simulate a situation you are used to

(*continued on next page*)

Table 5 (*continued*)

Best medium	Q: Could you elaborate on your preference?
Avatar	I think it would be best to keep it as simple as possible to not make it more cumbersome for students to interact with it, but simple text may be too simple. There is gain in having the feeling of interacting with some kind of avatar, and also having the possibility of getting information in forms other than simply text could increase engagement. Therefore I think the avatar is the best trade-off.
Chatbot (text-based)	I know it is fake, there is no point you trying to “Lie” to me by pretending someone is behind the screen

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